Our research and development efforts are focused on developing new and expanded features for our solutions, integrations with distributors and other software platforms, and improvements to our backend architecture. Research and development expenses consist primarily of personnel costs for employees and contractors, including share-based compensation expenses, and allocated costs of facilities and information technology, software tools, and product certification. We expense research and development costs as incurred, except for certain internal-use software development costs that we capitalize. We believe that continued investment in our products is important for our future growth, and we expect our research and development expenses to continue to increase in absolute dollars for the foreseeable future, although these expenses may fluctuate as a percentage of our total revenues from period to period depending on the timing of these expenses.

Sales and marketing expenses are the largest component of our operating expenses and consist primarily of personnel costs for employees and contractors directly associated with our sales and marketing activities including share-based compensation expenses, internet advertising fees, radio and billboard advertising, public relations, commissions paid to employees, resellers and other third parties, amortization of capitalized sales commissions, trade shows, travel expenses, credit card fees, marketing and promotional activities, amortization of acquired customer relationship intangibles, and allocated costs of facilities and information technology. We expect our sales and marketing expenses to continue to increase in absolute dollars for the foreseeable future as we expand our sales and marketing efforts domestically and internationally and continue to build our brand, although these expenses may fluctuate as a percentage of our total revenues from period to period depending on the timing of these expenses.

General and administrative expenses consist primarily of personnel costs, including share-based compensation expenses, for employees and contractors engaged in infrastructure and administrative activities to support the day-to-day operations of our business. Other significant components of general and administrative expenses include professional service fees, allocated costs of facilities and information technology, cost of compliance with certain government-imposed taxes, the costs of legal matters, business acquisition costs, and loss contingencies. We expect our general and administrative expenses to continue to increase in absolute dollars for the foreseeable future, although these expenses may fluctuate as a percentage of our total revenues from period to period, depending on the timing of these expenses.

Results of Operations

The following tables set forth selected consolidated statements of operations data and such data as a percentage of total revenues. The historical results presented below are not necessarily indicative of the results that may be expected for any future period (in thousands):

	Year ended December 31,		
	2019	2018	2017
Revenues			
Subscriptions	$ 817,811	$ 612,888	$ 465,254
Other	85,047	60,736	38,363
Total revenues	902,858	673,624	503,617
Cost of revenues			
Subscriptions	160,320	109,454	89,193
Other	70,723	47,675	32,078
Total cost of revenues	231,043	157,129	121,271
Gross profit	671,815	516,495	382,346
Operating expenses			
Research and development	136,363	101,042	75,148
Sales and marketing	439,100	329,116	240,223
General and administrative	142,027	102,773	72,313
Total operating expenses	717,490	532,931	387,684
Loss from operations	(45,675)	(16,436)	(5,338)
Other income (expense), net			
Interest expense	(20,512)	(16,102)	(99)
Other income, net	9,247	6,475	1,491
Other income (expense), net	(11,265)	(9,627)	1,392
Loss before income taxes	(56,940)	(26,063)	(3,946)
Provision for (benefit from) income taxes	(3,333)	140	258
Net loss	$ (53,607)	$ (26,203)	$ (4,204)